

Geometric Aspects of Sparse Multivariate Rational Function Interpolation with Application to solving Parametric Linear System.

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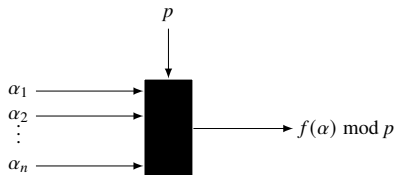
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Modular Black Box Definitions

Definition: A Modular Black Box $\mathcal{B}$ of a symbolic object over a field $\mathbb{F}$ is a computational procedure that takes a point $\alpha \in \mathbb{F}$ and a prime p as input, and returns the evaluation of the symbolic object at $\alpha \bmod p$ or FAIL if result is not defined.

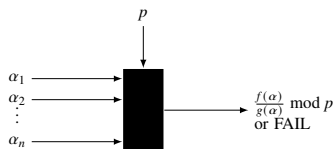
1. Multivariate Polynomials

Let $f \in \mathbb{F}[x_1, \dots, x_n]$. For $\alpha \in \mathbb{F}^n$.



2. Rational Functions

Let $h = \frac{f}{g} \in \mathbb{F}(x_1, \dots, x_n)$. For $\alpha \in \mathbb{F}^n$.



Outputs **FAIL** if $g(\alpha) \equiv 0 \pmod{p}$.

Modular Arithmetic and Reconstruction

- ① For a prime p , the elements of the field $\mathbb{F}_p$ are bounded by p , thus bounding operand sizes.
- ② Recover exact results:
 - ① use multiple primes $p_1, \dots, p_w$,
 - ② define $M = \prod_{i=1}^w p_i$.
- ③ We can recover a fraction $\frac{n}{d}$ from its modular images mod M using the Chinese remainder theorem (CRT) and Wang's rational number reconstruction (RNR) provided $M > 2|n| \cdot |d|$.

The Problem: Kaltofen Yang Sparse Multivariate Rational Function Interpolation

Let $h(x_1, \dots, x_n) = \frac{f(x_1, \dots, x_n)}{g(x_1, \dots, x_n)} \in \mathbb{F}(x_1, \dots, x_n)$ be a sparse multivariate rational function with black box $\mathcal{B} : (\mathbb{F}^n, p) \rightarrow \mathbb{F}$. We want to recover h by its black box evaluations.

① Multiple univariate rational function reconstructions

The Kaltofen Yang algorithm [5] constructs multiple univariate rational

functions $T_i(x) = \frac{\hat{f}_i(x)}{\hat{g}_i(x)} \in \mathbb{F}(x)$ by recovering $\mu \hat{f}_i(x), \mu \hat{g}_i(x) \in \mathbb{F}[x]$ for some scalar $\mu \in \mathbb{F}$ from the black box evaluations of h .

② Multivariate polynomial interpolation

Ben-Or Tiwari multivariate polynomial interpolation is used to recover $\mu f(x_1, \dots, x_n)$ from $\mu \hat{f}_i(x)$, and $\mu g(x_1, \dots, x_n)$ from $\mu \hat{g}_i(x)$, for some scalar $\mu \in \mathbb{F}$.

Example: Maximal Quotient Rational Function Reconstruction [7]

To recover $T(x) = \frac{3x^2+1}{x+7} \in \mathbb{Z}(x)$, with $p = 19$, we use the EEA on $m(x)$ and $u(x)$:

1. Computing $m(x), u(x)$

Let $\tau = 5 > \deg(\hat{f}) + \deg(\hat{g}) + 1$

Let $\alpha = \{1, 2, 3, 4, 5\}$.

$\mathcal{B}(\alpha) = \{10, 12, 18, 1, 0\}$.

Compute:

- $m(x) = \prod_{j=1}^{\tau=5} (x - \alpha_j) = x^5 + 4x^4 + 9x^3 + 3x^2 + 8x + 13$
- $u(x) = 17x^4 + 6x^3 + 16x^2 + 18x + 10$
as interpolating polynomial for the points $(\alpha_j, \mathcal{B}(\alpha_j))$.

2. Extended Euclidean Algorithm

Where $m(x)s_i(x) + u(x)t_i(x) = r_i(x)$ $1 \leq i \leq l$.

$$\implies u(x)t_i(x) \equiv r_i(x) \pmod{m(x)}$$

i	q_i	r_i	t_i
0	—	$m(x)$	0
1	$9x + 6$	$u(x)$	1
2	$5x^2 + 4x + 9$	$11x^2 + 10$	$10x + 13$
3	$9x + 7$	$16x + 15$	$7x^3 + 9x^2 + 10x + 17$

3. Recovery via Maximal Quotient

In row $i = 2$, $\deg(q_2)$ is maximal. Thus:

$$T(x) = \frac{r_2}{t_2} = \frac{11x^2 + 10}{10x + 13} \xrightarrow{\text{normalize}} \frac{10^{-1}r_2}{10^{-1}t_2} = \frac{3x^2 + 1}{x + 7}$$

Ben-Or Tiwari Multivariate Interpolation [2]

Let $f(x_1, \dots, x_n) = \sum_{k=1}^{\tau} b_k \mathcal{M}_k(x_1, \dots, x_n) \in \mathbb{F}[x_1, \dots, x_n]$ be the multivariate polynomial with black box $\mathcal{B} : (\mathbb{F}^n, p) \longrightarrow \mathbb{F}_p$ that we want to interpolate from its black box evaluations. Ben-Or Tiwari multivariate polynomial interpolation is a two-step algorithm that works when the black box of f is evaluated at increasing powers of a point α . Why? Because the sequence of evaluations $\{a_i\}_{i \geq 0} = \{f(\alpha^i)\}_{i \geq 0}$ forms a linear recurrence sequence of order τ . In practice, we take $\alpha = [2, 3, \dots, p_n]^T$.

- The first step of Ben-Or Tiwari algorithm finds the minimal characteristic polynomial of the sequence $\{a_i\}_{i \geq 0}$.

$$\Lambda(z) = z^{\tau} + c_{\tau-1}z^{\tau-1} + \dots + c_0 \in \mathbb{F}[z] = \prod_{k=1}^{\tau} (z - m_k). \quad (1)$$

where $m_k = \mathcal{M}_k(\alpha_1, \dots, \alpha_n)$ are the monomial evaluations at $\alpha \in \mathbb{F}^n$.

- In the second step, Ben-Or Tiwari algorithm recovers the coefficients by solving a transposed Vandermonde system.

Ben-Or Tiwari: Finding $\Lambda(Z)$

Gaussian Elimination of Hankel Matrix

- 1 Construct Matrix H_τ with a_i

$$H_\tau = \begin{bmatrix} a_0 & a_1 & \cdots & a_{\tau-1} \\ a_1 & a_2 & \cdots & a_\tau \\ \vdots & \vdots & \ddots & \vdots \\ a_{\tau-1} & a_\tau & \cdots & a_{2\tau-2} \end{bmatrix}$$

- 2 Solve Linear System

$$H_\tau \cdot c = -[a_\tau, \dots, a_{2\tau-1}]^T$$

$$\begin{bmatrix} a_0 & \cdots & a_{\tau-1} \\ \vdots & \ddots & \vdots \\ a_{\tau-1} & \cdots & a_{2\tau-2} \end{bmatrix} \begin{bmatrix} c_0 \\ \vdots \\ c_{\tau-1} \end{bmatrix} = - \begin{bmatrix} a_\tau \\ \vdots \\ a_{2\tau-1} \end{bmatrix}$$

Gives us the coefficients of $\Lambda(z)$ in $O(\tau^3)$ field operations.

Berlekamp-Massey Euclidean Algorithm

- 1 Given 2τ terms of a linear recurrence sequence

$$\{a_0, \dots, a_i = f(\alpha^i), \dots, a_{2\tau-1}\},$$

BMEA defines syndrome polynomial

$$S(z) \text{ using } a_i \text{ as } S(z) = \sum_{i=0}^{2\tau-1} a_i z^i$$

- 2 Solve Key Equation of LFSR

$S(z) \cdot \Lambda(z) \equiv \Omega(z) \pmod{z^{2\tau}}$ using EEA on $z^{2\tau}, S(z)$, until $\deg(\Omega) < \tau$. We know that for EEA with $z^{2\tau}, S(z)$ as inputs $z^{2\tau}s_i(z) + S(z)t_i(z) = r_i(z)$ holds. We get $\Lambda(z)$ as $t_i(z)$ in $O(\tau^2)$ field operations.

Ben-Or Tiwari: Finding Coefficients

To get the coefficients of the multivariate polynomial, we solve a transposed Vandermonde linear system:

$$V^T \mathbf{b} = \mathbf{y},$$

where

$$V^T = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ m_1 & m_2 & \cdots & m_\tau \\ \vdots & \vdots & \ddots & \vdots \\ m_1^{\tau-1} & m_2^{\tau-1} & \cdots & m_\tau^{\tau-1} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_\tau \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} f(1, 1, \dots, 1) \\ f(\alpha_1, \alpha_2, \dots, \alpha_n) \\ \vdots \\ f(\alpha_1^{\tau-1}, \alpha_2^{\tau-1}, \dots, \alpha_n^{\tau-1}) \end{bmatrix}$$

The transposed Vandermonde system can be solved using Zippel's Vandermonde solver in $O(\tau^2)$ field operations.

Kaltofen Yang ring isomorphism

Kaltofen and Yang define an isomorphism of rings that lets the algorithm construct univariate rational functions $T_i(x)$ from the black box evaluations of the multivariate rational function h and utilize Ben-Or Tiwari to recover $\mu f, \mu g \in \mathbb{F}[x_1, \dots, x_n]$ from $\mu \hat{f}, \mu \hat{g} \in \mathbb{F}[x]$.

$$\phi : \mathbb{F}[x_1, x_2, \dots, x_n, \sigma_1] \longrightarrow \mathbb{F}[x, \sigma_1, \dots, \sigma_n]$$

$$x_1 \longmapsto x,$$

$$x_j \longmapsto \beta_j (x - \sigma_1) + \sigma_j, \quad 2 \leq j \leq n.$$

$$\sigma_1 \longmapsto \sigma_1$$

The inverse map

$$\phi^{-1} : \mathbb{F}[x, \sigma_1, \dots, \sigma_n] \longrightarrow \mathbb{F}[x_1, x_2, \dots, x_n, \sigma_1]$$

$$\phi^{-1}(x) \longmapsto x_1,$$

$$\phi^{-1}(\sigma_1) \longmapsto \sigma_1$$

$$\phi^{-1}(\sigma_j) \longmapsto x_j - \beta_j (x_1 - \sigma_1) \quad 2 \leq j \leq n.$$

Expanding the coordinate transform $x_j \mapsto \beta_j(x - \sigma_1) + \sigma_j$ as $\beta_j x + (\sigma_j - \beta_j \sigma_1)$ where $2 \leq j \leq n$ and writing it as a vector gives us

$$\mathcal{L}(x) = \begin{bmatrix} 1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} x + \begin{bmatrix} 0 \\ \sigma_2 - \beta_2 \sigma_1 \\ \vdots \\ \sigma_n - \beta_n \sigma_1 \end{bmatrix}. \quad (2)$$

Geometric morphism and pullback

$$\Psi : \mathbb{A}^1 \longrightarrow \mathcal{L} \subset \mathbb{A}^n$$

$$a \longmapsto (a, \beta_2(a - \sigma_1) + \sigma_2, \dots, \beta_n(a - \sigma_1) + \sigma_n)$$

We define the inverse map

$$\Psi^{-1} : \mathcal{L} \longrightarrow \mathbb{A}^1$$

$$(a_1, \dots, a_n) \longmapsto (a_1)$$

$$\Psi \circ \Psi^{-1} = id_{\mathcal{L}} \quad \text{and} \quad \Psi^{-1} \circ \Psi = id_{\mathbb{A}^1} \implies \mathcal{L} \cong \mathbb{A}^1$$

The pullback will be

$$\Psi^* : \mathbb{F}[\mathcal{L}] \cong \mathbb{F}[x_1, \dots, x_n] / \mathcal{I}(\mathcal{L}) \longrightarrow \mathbb{F}[x]$$

Since $\mathcal{L} \cong \mathbb{A}^1$, $\mathbb{F}[\mathcal{L}] \cong \mathbb{F}[x]$. Since $\mathcal{I}(\mathcal{L})$ is a prime ideal, $\mathbb{F}[\mathcal{L}]$ will be an integral domain, thus $\Psi^* : \text{Frac}(\mathbb{F}[\mathcal{L}]) \cong \mathbb{F}(x)$.

Univariate rational function reconstruction from black box evaluations

$$h|_{\mathcal{L}_i} = \Psi^*(h) = \Psi^*\left(\frac{f}{g}\right) = \frac{\Psi^*(f)}{\Psi^*(g)} \quad (3)$$

$$\Psi^*(h(x_1, \dots, x_n)) = \frac{f(x, \beta_2(x - \sigma_1) + \sigma_2, \dots, \beta_n(x - \sigma_1) + \sigma_n)}{g(x, \beta_2(x - \sigma_1) + \sigma_2, \dots, \beta_n(x - \sigma_1) + \sigma_n)} = \frac{\hat{f}(x)}{\hat{g}(x)}.$$

Adjustment for Ben-Or Tiwari for requiring increasing powers of the point as input.

We modify Ψ slightly to $\Psi_i : \mathbb{A}^1 \longrightarrow \mathcal{L}_i$ such that $x \longmapsto (x, \beta_2(x - \sigma_1^i) + \sigma_2^i, \dots, \beta_n(x - \sigma_1^i) + \sigma_n^i)$

$$\Psi_i^*(h(\sigma_1^i, \dots, \sigma_n^i)) = \frac{f(\sigma_1^i, \dots, \sigma_n^i)}{g(\sigma_1^i, \dots, \sigma_n^i)} = \frac{\hat{f}_i(\sigma_1^i)}{\hat{g}_i(\sigma_1^i)} = T_i(\sigma_1^i). \quad (4)$$

Kaltofen Yang algorithm

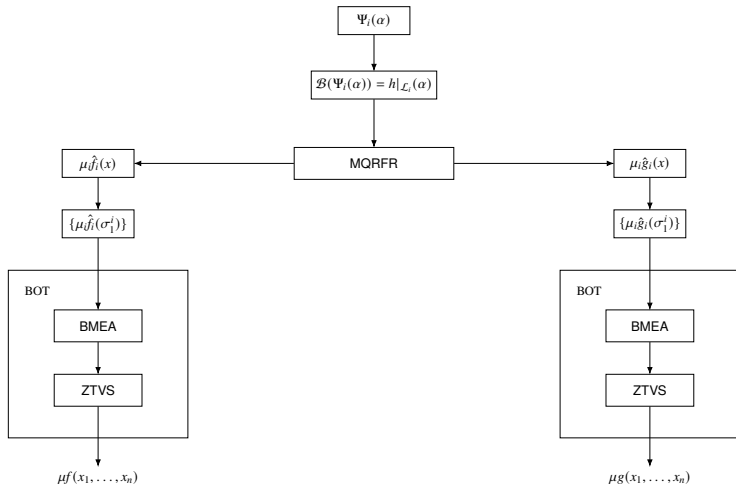
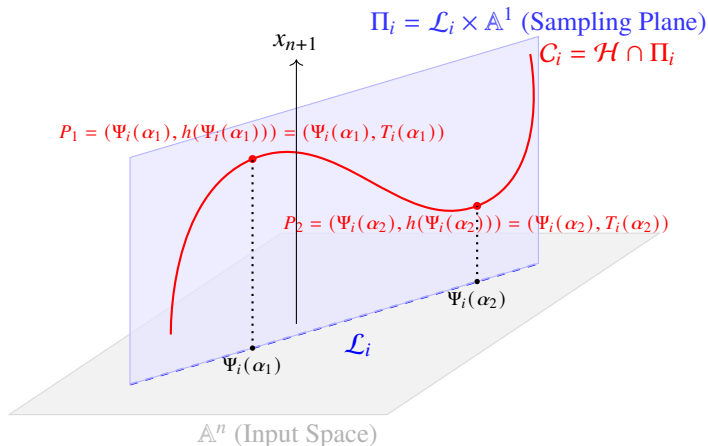


Figure: High level overview of Kaltofen Yang algorithm

Sampling plane over $\mathcal{L}_i$ and intersecting curve C_i

The graph of rational function h $\Gamma_h = \{(x_1, \dots, x_n, h(x_1, \dots, x_n))\}$ will be a hypersurface $\mathcal{H} \subset \mathbb{A}^n \times \mathbb{A}^1$.



Example: Affine lines and Black box evaluations

Consider $h(x, y) = \frac{x + 5y}{xy + 3} \in \mathbb{R}(x, y)$

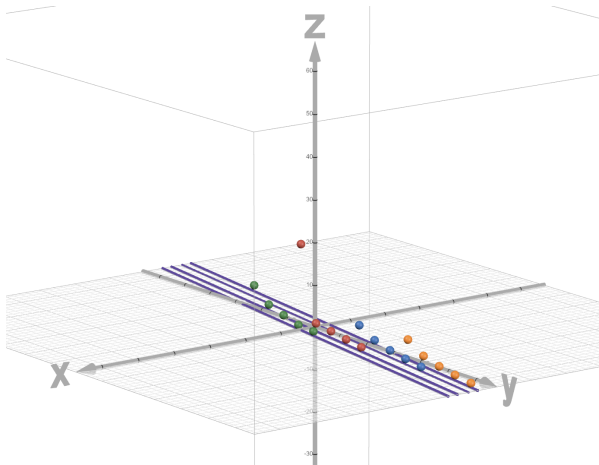
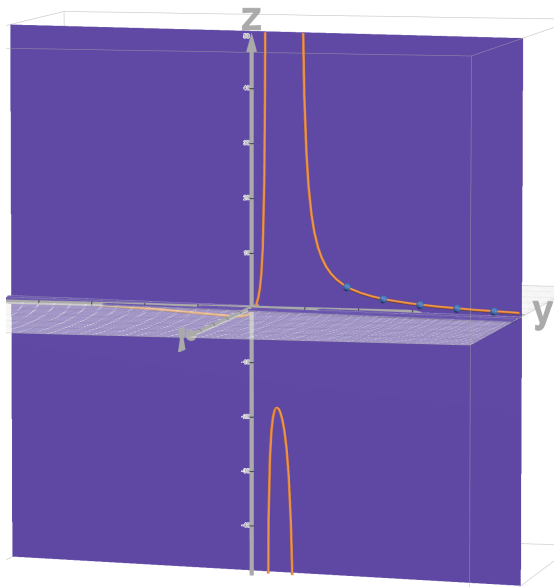
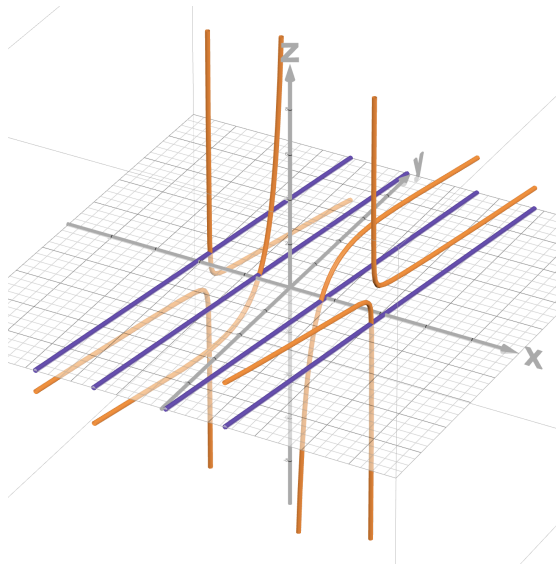


Figure: Affine lines $\mathcal{L}_i$ and the points $(\Psi_i(\alpha_j), h(\Psi_i(\alpha_j)))$

Sampling plane and intersecting curve

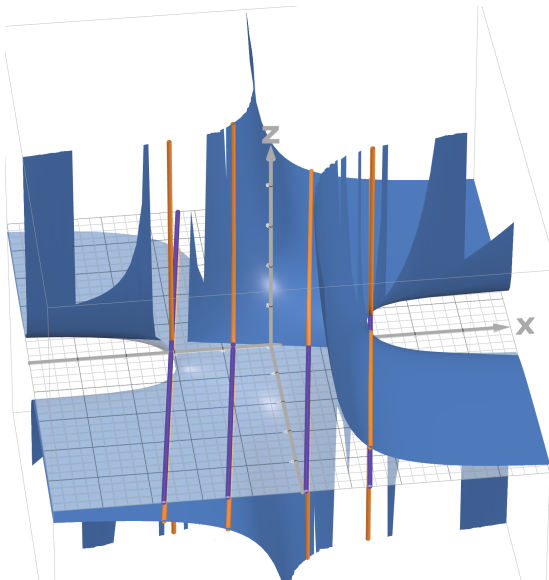


All intersecting curves



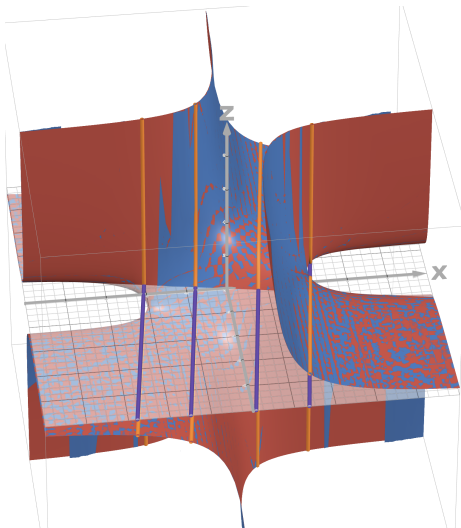
Hypersurface $\mathcal{H}$

The graph of Γ_h defined as $(x, y, h(x, y))$ is a hypersurface $\mathcal{H} \subset \mathbb{A}^2 \times \mathbb{A}^1$.



Zarisky closure

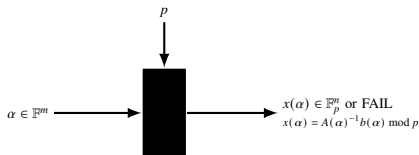
The Zarisky closure of Γ_h where $h = \frac{f}{g} \in \mathbb{F}(x_1, \dots, x_n)$ is $V(x_{n+1}g - f)$. Here it will be $V(z(xy + 3) - (x + 5y))$.



Application: Solving Parametric Linear Systems

The Problem: Solve $Ax = b$ where A and b contain parameters $y_1, \dots, y_m$.

- ❶ Where $A \in \mathbb{F}[y_1, \dots, y_m]^{n \times n}$ and $b \in \mathbb{F}[y_1, \dots, y_m]^n$.
- ❷ By Cramer's rule, the solution of such systems are rational functions:
$$x_i = \frac{f_i}{g_i} = \frac{\det(A_i)}{\det(A)} \in \mathbb{F}(y_1, \dots, y_m).$$
- ❸ The modular black box $\mathcal{B}$ for the parametric system $Ax = b$ is a program that
 - takes a point α in the parameter space $\mathbb{F}^m$, and a prime p as input,
 - evaluates $A(\alpha) \in \mathbb{F}^{n \times n}, b(\alpha) \in \mathbb{F}^n$,
 - solves the system at point α using Gaussian elimination and
 - returns the solution vector $x(\alpha) \in \mathbb{F}^n = A(\alpha)^{-1} \cdot b(\alpha) \pmod p$ or FAIL.



We use the Kaltofen Yang algorithm to reconstruct the solution vector $x \in \mathbb{F}(y_1, \dots, y_m)^n$ by interpolating each component $x_i(\alpha) = \frac{f_i(\alpha)}{g_i(\alpha)}$.

Rational B spline system

Consider the system $A\mathbf{x} = b$, where $A \in \mathbb{Z}[y_1, \dots, y_5]^{21 \times 21}$ and $b \in \mathbb{Z}^{21}$. The solution vector $\mathbf{x} \in \mathbb{Z}(y_1, \dots, y_5)^{21}$.

$$x_7 + x_{12} = 1 \quad (1)$$

$$x_8 + x_{13} = 1 \quad (2)$$

$$x_{21} + x_6 + x_{11} = 1 \quad (3)$$

$$x_1 y_1 + x_1 - x_2 = 0 \quad (4)$$

$$x_{11} y_3 + x_{11} - x_{12} = 0 \quad (5)$$

$$x_{16} y_5 - x_{17} y_5 - x_{17} = 0 \quad (6)$$

$$-x_{20} y_3 + x_{21} y_3 + x_{21} = 0 \quad (7)$$

$$x_3 y_2 + x_3 - x_4 = 0 \quad (8)$$

$$-x_8 y_4 + x_9 y_3 + x_9 = 0 \quad (9)$$

$$2x_1 y_1^2 - 2x_1 - 2x_{10} + 4x_2 = 0 \quad (10)$$

$$-x_{10} y_2 + x_{18} y_2 + x_{18} - x_{19} = 0 \quad (11)$$

$$2x_{11} y_3^2 - 2x_{11} + 4x_{12} - 2x_{13} = 0 \quad (12)$$

$$-x_{13} y_4 + x_{14} y_4 + x_{14} - x_{15} = 0 \quad (13)$$

$$2x_{15} y_5^2 - 4x_{16} y_5^2 + 2x_{17} y_5^2 - 2x_{17} = 0 \quad (14)$$

$$2x_{19} y_3^2 - 4x_{20} y_3^2 + 2x_{21} y_3^2 - 2x_{21} = 0 \quad (15)$$

$$2x_3 y_2^2 - 2x_3 + 4x_4 - 2x_5 = 0 \quad (16)$$

$$-x_5 y_3 + x_6 y_3 + x_6 - x_7 = 0 \quad (17)$$

$$2x_7 y_4^2 - 4x_8 y_4^2 + 2x_9 y_4^2 - 2x_9 = 0 \quad (18)$$

$$-4x_{10} y_2^2 + 2x_{18} y_2^2 + 2x_{20} y_2^2$$

$$-2x_{18} + 4x_{19} - 2x_{20} = 0 \quad (19)$$

$$2x_{12} y_4^2 - 4x_{13} y_4^2 + 2x_{14} y_4^2$$

$$-2x_{14} + 4x_{15} - 2x_{16} = 0 \quad (20)$$

$$2x_4 y_3^2 - 4x_5 y_3^2 + 2x_6 y_3^2$$

$$-2x_6 + 4x_7 - 2x_8 = 0 \quad (21)$$

Solution System (Part 1: $x_1 - x_{13}$)

It took 973 black box probes to solve this system using prime $p = 2^{31} - 1$.

$$x_1 = \frac{1}{y_1^2 y_2 + y_1^2 + y_1 y_2 + 2y_1 + 1}$$

$$x_2 = \frac{1}{y_1 y_2 + y_1 + 1}$$

$$x_3 = \frac{1}{y_2^2 y_3 + y_2^2 + y_2 y_3 + 2y_2 + 1}$$

$$x_4 = \frac{1}{y_2 y_3 + y_2 + 1}$$

$$x_5 = \frac{y_2 + 1}{y_2 y_3 + y_2 + 1}$$

$$x_7 = \frac{2y_3^2 y_4 - y_3 y_4^2 + 2y_3 y_4 + y_3}{2y_3^2 y_4 - y_3 y_4^2 + 3y_3 y_4 - y_4^2 + y_3 + y_4 + 1}$$

$$x_8 = \frac{y_4 y_3}{2y_3 y_4 - y_4^2 + y_4 + 1}$$

$$x_9 = \frac{y_3 y_4^2}{2y_3^2 y_4 - y_3 y_4^2 + 3y_3 y_4 - y_4^2 + y_3 + y_4 + 1}$$

$$x_{10} = \frac{y_1 + 1}{y_1 y_2 + y_1 + 1}$$

$$x_{11} = \frac{y_3 y_4 - y_4^2 + y_4 + 1}{2y_3^2 y_4 - y_3 y_4^2 + 5y_3^2 y_4 - 2y_3 y_4^2 + y_3^2 + 4y_3 y_4 + 2y_3 - y_4^2 + y_4 + 1}$$

$$x_{12} = \frac{y_3 y_4 - y_4^2 + y_4 + 1}{2y_3^2 y_4 - y_3 y_4^2 + 3y_3 y_4 - y_4^2 + y_3 + y_4 + 1}$$

$$x_{13} = \frac{y_3 y_4 - y_4^2 + y_4 + 1}{2y_3 y_4 - y_4^2 + y_4 + 1}$$

$$x_6 = \frac{4y_2^3 y_3 y_4 - 2y_2^2 y_3^2 y_4 + 7y_2 y_3^2 y_4 - 2y_2 y_3 y_4^2 + 2y_3^3 y_4 - y_2^2 y_4^2 + 2y_2 y_3^2 + 3y_2 y_3 y_4 + 5y_3^2 y_4 - 2y_3 y_4^2 + 2y_2 y_3 + y_3^2 + 3y_3 y_4 + 2y_3}{2y_2^4 y_3 y_4 - y_2^3 y_3^2 y_4 + 7y_2^2 y_3^2 y_4 - 3y_2 y_3^2 y_4^2 + y_2 y_3^3 + 9y_2^2 y_3^2 y_4 + 3y_2 y_3^2 - 3y_2 y_3 y_4^2 + 5y_2 y_3 y_4 + 3y_2 y_3 - y_2 y_4^2 + y_2 y_4 + y_2 + 2y_3^3 y_4 - y_2^2 y_4^2 + 5y_3^2 y_4 - 2y_3 y_4^2 + y_3^2 + 4y_3 y_4 + 2y_3 - y_4^2 + y_4 + 1}$$

Solution System (Part 2: $x_{14} - x_{21}$)

$$x_{14} = \frac{2y_3^2y_4^3y_5 - 2y_3y_4^4y_5 + 2y_3^2y_4^3 + y_3^2y_4^2y_5 - 2y_3y_4^4 + 2y_3y_4^3y_5 - y_4^4y_5 + 2y_3^2y_4^2 + y_3y_4^3 + 4y_3y_4^2y_5 - y_4^4 + 6y_3y_4^2 + y_3y_4y_5 - y_4^3 + 2y_4^2y_5 + 2y_3y_4 + 3y_4^2 + y_4y_5 + 2y_4}{2y_3^2y_4^3y_5 - y_3y_4^4y_5 + 2y_3^2y_4^3 + 2y_3^2y_4^2y_5 - y_3y_4^4 + 2y_3y_4^3y_5 - y_4^4y_5 + 4y_3^2y_4^2 + y_3y_4^3 + 4y_3y_4^2y_5 - y_4^4 + 2y_3^2y_4 + 6y_3y_4^2 + y_3y_4y_5 - y_4^3 + 2y_4^2y_5 + 5y_3y_4 + 2y_4^2 + y_4y_5 + y_3 + 3y_4 + 1}$$

$$x_{15} = \frac{y_3^2y_4^3y_5 - y_3y_4^4y_5 + y_3^2y_4^2y_5 + 3y_3y_4^2y_5 - y_4^3y_5 + y_3y_4y_5 + y_4^2y_5 + y_4y_5 + y_3^2y_4^3 - y_3y_4^4 + y_3^2y_4^2 + 3y_3y_4^2 - y_4^3 + y_3y_4 + y_4^2 + y_4}{2y_3^2y_4^3y_5 - y_3y_4^4y_5 + 2y_3^2y_4^2 + 3y_3y_4^2y_5 - y_4^3y_5 + 2y_3^2y_4 + y_3y_4y_5 + y_4^2y_5 + y_3y_4 + y_4y_5 - y_3y_4^2 + 3y_3y_4 - y_4^2 + y_3 + y_4 + y_5 + 1}$$

$$x_{16} = \frac{y_3^2y_4^3y_5 - y_3y_4^4y_5 + y_3^2y_4^2y_5 + 3y_3y_4^2y_5 - y_4^3y_5 + y_3y_4y_5 + y_4^2y_5 + y_4y_5}{2y_3^2y_4^3y_5 - y_3y_4^4y_5 + 2y_3^2y_4^2 + 3y_3y_4^2y_5 - y_4^3y_5 + 2y_3^2y_4 + y_3y_4y_5 + y_4^2y_5 + y_3y_4 + y_4y_5 - y_3y_4^2 + 3y_3y_4 - y_4^2 + y_3 + y_4 + y_5 + 1}$$

$$x_{17} = \frac{y_3^3y_4^3y_5^2 - y_3y_4^4y_5^2 + y_3^2y_4^2y_5^2 + 3y_3y_4^2y_5^2 - y_4^3y_5^2 + y_3y_4y_5^2 + y_4^2y_5^2 + y_4y_5^2}{2y_3^3y_4^3y_5^2 - y_3y_4^4y_5^2 + 3y_3y_4^2y_5^2 - y_4^3y_5^2 + y_3y_4y_5^2 + y_4^2y_5^2 + y_4y_5^2 + 4y_3^2y_4^2y_5 - 2y_3y_4^3y_5 + 6y_3y_4^2y_5 - 2y_4^3y_5 + 2y_3^2y_4y_5 + 3y_3y_4y_5 + 3y_4^2y_5 + y_3y_5 + 2y_4y_5 + y_5 + 2y_3^2y_4 - y_3y_4^3 + 3y_3y_4^2 - y_4^3 + 2y_3^2y_4 + y_3y_4 + y_4^2 + y_3 + 2y_4 + 1}$$

$$x_{18} = \frac{2y_1y_2^2y_3 + 2y_1y_2^2 + y_1y_2y_3 + y_2^2y_3 + 2y_1y_2 + y_2^2 + y_2y_3 + 2y_2}{y_1y_2^3y_3 + y_1y_2^2 + 2y_1y_2^2y_3 + 3y_1y_2^2 + 2y_1y_2y_3 + 3y_1y_2 + y_1 + y_2^2y_3 + y_2^2 + y_2y_3 + 2y_2 + 1}$$

$$x_{19} = \frac{y_2y_3 + y_2}{y_2y_3 + y_2 + 1}$$

$$x_{20} = \frac{y_2y_3}{y_2y_3 + y_2 + 1}$$

$$x_{21} = \frac{y_2y_3^2}{y_2y_3^2 + 2y_2y_3 + y_2 + y_3 + 1}$$

Comparing results with Bareiss/Edmonds/Lipson [4]

Let $\tilde{x} = \tilde{x}_i = \frac{N_i}{D_i}$ for $1 \leq i \leq 21$ be the solution of the rational B spline system computed using Bareiss/Edmonds/Lipson algorithm[1, 3, 6]. Let $\#N_i$ be the number of terms in the numerator N_i and $\#D_i$ be the number of terms in the denominator D_i of $\tilde{x}$.

Whereas $x_i = \frac{f_i}{g_i}$ for $1 \leq i \leq 21$ are interpolated in simplified form using the Kaltopen Yang algorithm .

Table: Number of terms in N_i , D_i , f_i and g_i .

	1	2	3	4	5	6	7	8	9	10	11
$\#N_i$	586	3,490	7,410	4,940	7,072	11,793	12,802	11,211	9,620	1,172	1,197
$\#D_i$	2	36	153	153	432	672	672	672	672	3	6
$\#f_i$	1	1	1	1	2	14	4	1	1	2	4
$\#g_i$	5	3	5	3	3	23	7	4	7	3	10

	12	13	14	15	16	17	18	19	20	21
$\#N_i$	1,827	2,142	1,666	2,072	1,320	1,320	2,650	2,543	3,971	5,675
$\#D_i$	9	9	9	9	9	18	18	27	36	117
$\#f_i$	4	4	19	16	8	8	8	2	1	1
$\#g_i$	7	4	22	16	16	26	12	3	3	5

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